

Science Activity Sheet

Quarter 3 – MELC 1

Week 2

Effects of Friction and Gravity on Movement of Objects



REGION VI – WESTERN VISAYAS

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Science 6

Activity Sheet No. 2- Effects of Friction on Movement of Objects

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Development Team of Science Activity Sheet

Writers: Nesa Marie Kristelle Tindoc, Edna Rose P. Gueco

Illustrator: Raymond Gayatin, Edna Rose P. Gueco

Layout Artist: Nesa Marie Kristelle Tindoc, Edna Rose P. Gueco

Schools Division Quality Assurance Team:

Oswald P. Acebedo
Marielle Grace A. De Oca
Ellen G. De la Cruz

Division of Bacolod City Management Team

Gladys Amylaine D. Sales
Michell L. Acoyong
Janalyn B. Navarro
Edna Rose P. Gueco
Ellen G. De la Cruz

Regional Management Team

Ramir B. Uytico
Pedro T. Escobarte Jr.
Elena P. Gonzaga
Donald T. Genine
Rovel R. Salcedo
Moonyeen C. Rivera
Anita S. Gubalane
Minda L. Soldevilla
Daisy L. Lopez
Joseph M. Pagalaran

Introductory Message

Welcome to **Science 6!**

The **Learning Activity Sheet** is a product of the collaborative efforts of the Schools Division of Bacolod City and DepEd Regional Office VI - Western Visayas through the Curriculum and Learning Management Division (CLMD). This is developed to guide the learning facilitators (teachers, parents and responsible adults) in helping the learners meet the standards set by the K to 12 Basic Education Curriculum.

The **Learning Activity Sheet** is self-directed instructional materials aimed to guide the learners in accomplishing activities at their own pace and time using the contextualized resources in the community. This will also assist the learners in acquiring the lifelong learning skills, knowledge and attitudes for productivity and employment.

For learning facilitator:

The **Science 6 Activity Sheet** will help you facilitate the leaching-learning activities specified in each Most Essential Learning Competency (MELC) with minimal or no face-to-face encounter between you and learner. This will be made available to the learners with the references/links to ease the independent learning.

For the learner:

The **Science 6 Activity Sheet** is developed to help you continue learning even if you are not in school. This learning material provides you with meaningful and engaging activities for independent learning. Being an active learner, carefully read and understand the instructions then perform the activities and answer the assessments. This will be returned to your facilitator on the agreed schedule.

Name of Learner: _____

Grade and Section: _____ Date: _____

SCIENCE 6 ACTIVITY SHEET NO. 2

Effects of Friction on Movement of Objects

I. Learning Competency with Code

Infer how friction and gravity affect the movements of different objects
(S6FE-IIIac-1).

II. Background Information for Learners

Friction is a force that occurs between the surfaces of two objects rubbing against each other. It may vary depending on the type of surface an object comes in contact with. With this, friction brings advantages and disadvantages to moving and non-moving objects. Learn and discover how friction and gravity affect moving objects.

III. Activity Proper

A. Activity 1: Rolling and Sliding

Directions: In this activity, you will observe how friction affects the movement of objects and identify the factors affecting them. Write your answers on a separate sheet or in your science notebook.

Materials: You will need a small box (matchbox) and a big marble.

Procedure:

1. Perform this activity on a floor or on a long table.
2. Put a mark on the floor or table. Place the big marble on the mark.
3. Push the marble slightly, as shown in Figure 1 below. Mark the point where the marble stopped.
4. Replace the marble with a small box (matchbox). Place it on the same mark where the marble was positioned earlier.
5. Apply the same amount of push to the box, as shown in Figure 2. Put a mark on the point where the small box stopped.



Figure 1: Boy pushing a marble.

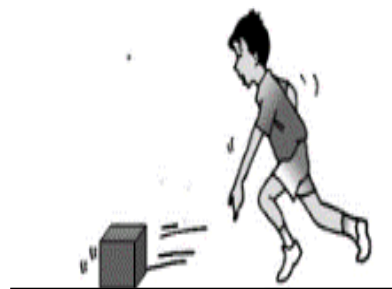


Figure 2: Boy pushing a matchbox.

Fill in the table.

Based on your observation, compare the following:	Write Marble or Matchbox
1. Which object has the longer distance traveled?	
Which object has the shorter distance traveled?	
2. Which object has a bigger surface area?	
Which object has a smaller surface area?	
3. Which object was acted by greater friction?	
Which object was acted by lesser friction?	

Guide Questions:

Based on the activity, answer the questions below.

1. Why do the marble and the matchbox travel at different distances?

2. How does friction affect the movement of objects such as marble and a matchbox?

3. What causes the difference in the distance travelled by the marble and matchbox?

4. In this activity, what is the factor affecting friction? _____

Activity 2: Rough and Smooth

Directions: Prepare the following materials and follow the procedures provided to answer the given questions. Write your answers on a separate sheet or in your science notebook.

Materials: You will need any round object like a marble, a ball, or a tomato and a piece of cloth (like a towel).

Procedure:

1. Perform this activity on a long table or on the floor.
2. Set up all the materials, as shown in Figure 3.
3. Cover one side of the table with a cloth (like a towel). The covered portion will represent the rough surface and the uncovered portion for a smooth surface.
4. Place the big marble at the starting line of the smooth surface.
5. Push the marble slightly. Mark the point where the marble stopped.
6. Do the same for the rough surface this time.

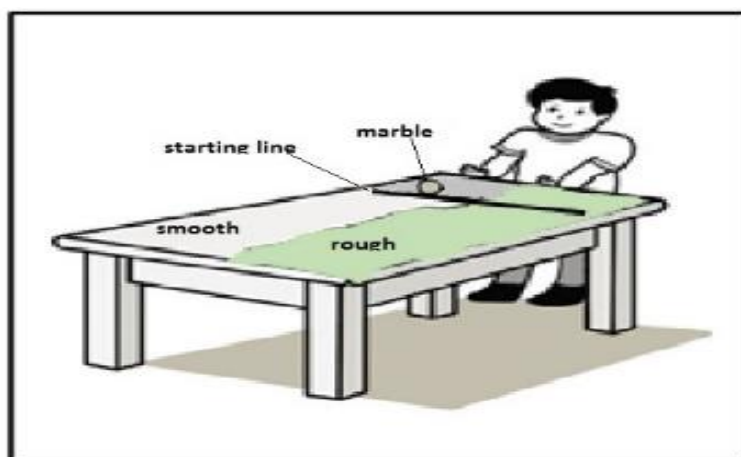


Figure 3: Boy tried to roll the marble on smooth and rough.

Fill in the table below.

Questions	Write Smooth or Rough
1. On which surface does the ball traveled a longer distance?	
2. On which surface does the ball traveled a shorter distance?	
3. Which surface has greater friction?	
4. Which surface has lesser friction?	

Guide Questions

Answer the questions below.

5. How does friction differ in a smooth surface from a rough surface?

6. How does friction affect the movement of the object on smooth and rough surfaces?

7. In this activity, what is the factor affecting friction? _____

8. How does friction affect the movement of objects?

Factors that Affects the Amount of Friction

The amount of friction depends on the following factors:

1. Mass of the Object

Object with greater mass has greater weight (W) that presses the surface of the ground downward. A Normal force (N) is the force exerted by the ground back on the object. It has the same amount with that of the weight. Since Normal force and weight are equal, then the bigger object has greater normal force than the smaller object (Figure 4). Friction is greater in an object with greater mass.

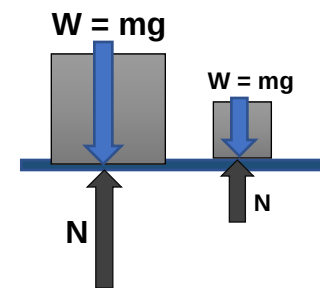


Figure 4. Weight is equal to Normal Force

2. Area of Contact

Object with a bigger surface in contact with another object produces greater friction.

Take a look at Figure 5 below. It may be very difficult to move the box to the right because of the greater friction that acts opposite the direction of motion (Figure 5a). What will you do to move the box easily? The use of rollers reduces friction when moving a heavy box (Figure 5b) that makes it easier to move. In this case, the sliding friction offers greater resistance than the rolling friction.

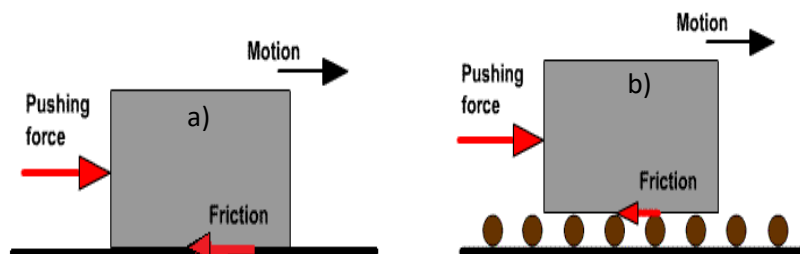


Figure 5. Example of sliding friction (a) and rolling friction (b)

3. Roughness of the surfaces in contact - rough surface offers greater friction than the smooth surface.

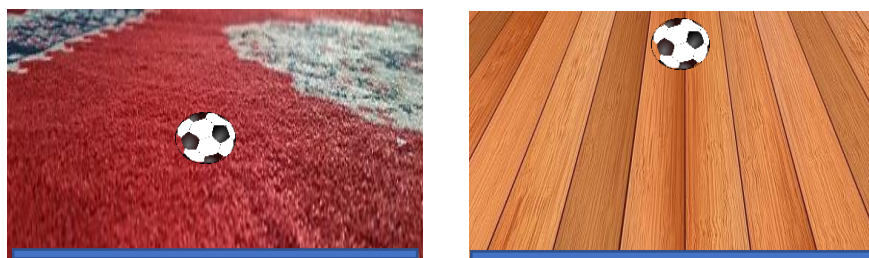


Figure 6. Ball rolled on the carpeted floor (a) and on the smooth plain wooden floor (b).

Figure 6 showed that the ball rolled farther on the plain smooth floor than the carpeted floor. The carpeted floor has a rough surface compared with the plain smooth wooden floor. A rough surface has greater friction than a smooth surface.

Fluids such as air and water resist movements of objects like falling stones, airplanes, ships and fishes.

For falling objects, air friction slows down its movement. For example, skydivers who jump off from the airplane fall at a faster rate. But once they open their parachutes, they descend or fall slowly (Figure 7).



Figure 7. Parachute slows down the speed of fall.

Air friction acts opposite the direction of motion of falling objects as shown on figure 8. The bigger or wider the surface area of falling objects, the greater the air friction. The smaller the surface area of falling objects, the air friction is lesser regardless of object's mass.

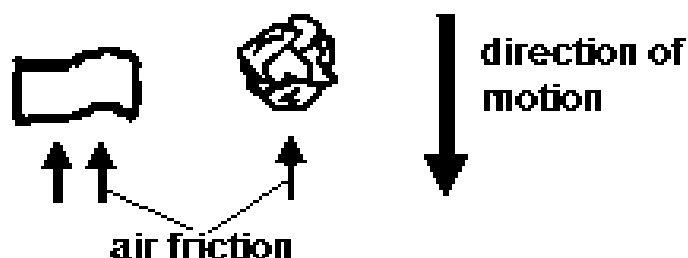


Figure 8: Air friction acting on the paper.

IV. Reflection

How does friction affect your daily activities?

V. Key Answer

Activity 1: Rolling and Sliding

Fill in the table

1. marble
2. matchbox
3. matchbox
marble
3. matchbox

Guide Questions

Possible Answers

1. The ball and the box travel different distances because of the different shape/surface area.
2. An object with a bigger surface area, has bigger friction while an object with a smaller surface area has a smaller friction.
3. Friction makes objects' motion slowdown. It can also stops moving objects.
4. Surface area

Activity 2: Rough and Smooth

1. Fill in the table

1. Smooth
2. Rough
3. Rough
4. Smooth

Guide Questions

Possible Answers

5. Friction is smaller in smooth surface, while friction is bigger in rough surface.
6. Object moves faster on smooth surface. Object moves slower on rough surface.
7. Texture
8. Friction affects the movement of the ball by acting to resist the sliding between the two touching surfaces. The rolling ball will move until friction will slow it down and stop its movement.

Reflection:

Friction affects our life in almost everything that we do. By simple picking up something using our bare hands, friction is acting on the surface of our skin and the thing that we grip. Walking and riding to our destination is also possible because friction prevents the car or our shoes from slipping or sliding. All things that touch each other's surfaces are affected by friction.

VI. Other References

Book

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